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Vision language models for bridge inspections: A review of applications in image-based damage documentation

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Abstract

Automating bridge inspections requires more than detecting individual damage instances. It demands systems capable of describing, contextualizing, and interpreting damage in an inspection-relevant manner. Conventional computer vision approaches, such as object detection and segmentation, primarily address visual recognition tasks and are constrained by predefined damage classes. Recent advances in vision-language models (VLMs), particularly foundation models with few-shot and zero-shot learning capabilities, offer new opportunities by enabling semantically rich descriptions and reasoning over visual scenes. This paper presents an emerging-trend review of VLM applications in bridge inspection, with a specific focus on their ability to satisfy information requirements derived from inspection practice. The synthesis identifies promising research directions, such as image captioning, damage cause inference, and human–AI collaboration, while also revealing that existing studies often leverage VLMs only superficially, frequently limiting them to classification or rudimentary captioning. Persistent challenges remain, including domain adaptation, data quality, image scale, and the absence of multimodal benchmarks. To address these limitations, the paper proposes a conceptual, inspection-oriented VL benchmark framework and outlines research directions required to move beyond damage classification toward operational multimodal systems that support documentation and interpretation.

KEYWORDS

CLIP, crack, image caption, VQA

1 | INTRODUCTION

Automating the generation of damage documentation is a central objective in the digitalization of bridge inspections. Yet structural damage is often visually complex: it may be occluded, ambiguous, or embedded in semantically dense scenes that require interpretation rather than mere detection. Figure 1 illustrates such cases, where visual symptoms, environmental context, and co-occurring damages must be understood jointly to produce adequate documentation.

Recent advances in vision-language models (VLMs), particularly vision-language (VL) foundation models, offer new opportunities for addressing this challenge. By

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FIGURE 1 Example images of complex damage scenes illustrating ambiguity, damage symptoms, labeling challenges, and the need for contextual interpretation beyond simple detection. Source of damage images (a), (b), and (d): InfraInspecta GmbH. Source of image (c): Kempen Krause Ingenieure GmbH. (a) A crack is fully obscured by moss growth, making crack segmentation challenging. (b) Co-occurring symptoms such as moisture ingress, moss growth, efflorescence, and corrosion products clearly indicate a leaking structure. In this case, conventional detection methods are redundant, as (a) the primary damage, the leakage itself, cannot be directly detected, and (b) detailed detection of the symptoms is unnecessary. (c) A leaking ceiling joint has caused moisture ingress, efflorescence, and corrosion of the reinforcement. The corroded reinforcement is visible only through the pattern of rust discoloration. Image detection models can neither detect nor infer the presence of hidden corroded rebar from such surface symptoms. (d) Corrosion on a handrail has produced a parallel rust stain on the ground. This stain does not indicate subsurface corrosion but results from rainwater transporting corrosion particles to the ground, illustrating a potential false positive for detection models.

integrating computer vision (CV) and natural language processing (NLP), VLMs can generate descriptive text, answer questions, and link visual evidence to semantic concepts, enabling forms of contextual reasoning and interpretation that conventional CV methods cannot provide.

To date, image-based machine learning (ML) approaches such as classification, object detection, and semantic segmentation have been widely applied to bridge inspection, with particular emphasis on crack detection.[1,2] While these methods have improved automated damage identification, they remain limited to predefined classes and cannot capture relationships between damages, components, and broader structural context. Consequently, methods capable of describing, summarizing, and interpreting damage scenes are increasingly required.

Structural inspection research also faces systemic constraints. Large, high-quality benchmark datasets remain scarce, and existing datasets cover only a narrow subset of damage types despite a broad set defined in bridge

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management systems (BMSs). As a result, most studies continue to focus on a small set of recurring damages and deal with them within the same task levels (bounding boxes or segmentation), even though different damage phenomena require different forms of representation, ranging from pixel-level detection to higher-level textual description.

Given these constraints, the application of VLMs to bridge inspection emerges as a promising research area. To date, only 13 peer-reviewed studies address this specific research intersection. Although several review papers have examined CV-based approaches for damage detection, none has provided a dedicated and domain-specific analysis of VLM research in bridge inspections. For instance, Pan et al.[3] surveyed recent CV developments for damage detection and identified it as a future research direction yet did not provide an in-depth review of existing VLM-based studies. Amirkhani et al.[2] reviewed commonly used concrete bridge damages and summarized applied ML approaches without considering VLMs. Zhang et al.,[1] in turn, analyzed crack detection literature with a focus on learning paradigms, mentioning VLMs only briefly. Despite these contributions, the literature lacks a systematic review of how and why VLMs are currently applied to bridge inspection tasks.

To address this gap, this paper presents an emerging-trend review of VLM applications for bridge inspection and damage documentation. Given the early stage of the field, this review establishes a structured overview of the nascent research landscape, identifies methodological patterns, and analyzes studies against inspection-derived information requirements. Building on this analysis, key research challenges are discussed, directions for advancing the use of VLMs in automated bridge inspection are derived, and a conceptual benchmark framework for inspection-oriented VL applications is proposed. Specifically, this review addresses the following research questions:

1. How have VLMs been applied to damage documentation and bridge inspection to date?
2. What motivates their use, and which limitations remain?
3. What information requirements do bridge inspections entail, and how are they reflected in current VLM-based studies?
4. What requirements must a VL benchmark for bridge inspection fulfill?

The remainder of this paper is structured as follows: Section 2 provides an overview of the evolution of VLMs and introduces key model families relevant to this review. Section 3 describes the literature search methodology. Section 4 presents and analyzes existing studies applying VLMs within the context of bridge inspection. Section 5 discusses the reviewed literature, with particular emphasis on information requirements and methodological gaps. Based on this discussion, Section 6 derives implications for the design of inspection-oriented VL benchmarks. Section 7 outlines future research directions, followed by conclusions in Section 8.

2 | VISION-LANGUAGE MODELS

Integrating language and vision opens great potential automating diverse tasks. The family of ML models that jointly process visual and textual information is commonly referred to as VLMs. A broader term, which is multimodal models, also encompasses systems that integrate additional modalities beyond text and images, such as audio. VLMs typically take images and text as input and produce textual outputs, such as descriptions and answers. Beyond text generation, some VLMs also provide spatial outputs (e.g., bounding boxes or segmentation masks), enabling concrete linking of text to image regions.

The foundations of VLMs were laid by statistical methods to provide captions to images (e.g., reference 4). With the rise of deep learning in the mid-2010s, the capabilities of captioning models improved significantly (e.g., references 5,6). In parallel, additional VL tasks were defined, including video captioning,[7] visual question answering (VQA) [Correction added on 30 July 2026, after first online publication: Citation to reference 36 was inadvertently added and has been removed in this version.],[8] visual grounding (also called phrase grounding),[9] visual storytelling,[10] and visual dialog.[11] Although additional tasks have been defined in the literature, the overview provided here is limited to those most widely addressed.

In VQA, models answer natural-language questions about an image and are trained on triplets of image, question, and answer. Phrase grounding links textual expressions to specific image regions, typically by using bounding boxes or segmentation masks. Visual storytelling generates cohesive narrative texts from image sequences. Visual dialog enables conversation about visual content.

The first ML models developed for VL tasks were built on convolutional neural networks (CNNs), recurrent neural networks (RNNs), and long short-term memory (LSTMs) networks. In general, VLM architectures first extract image and text features separately, then fuse them into a joint multimodal embedding space that is processed by language layers to generate text outputs. Two major developments in the ML field significantly advanced VLMs. The first was the introduction of the Transformer architecture.[12] Transformers became the

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standard for text embeddings and the layers after the fusion of image-text features. Until the introduction of vision transformers (ViT),[13] images were processed through object detection models, such as Faster R-CNN,[14] which provided visual features based on image regions. Representative region-based models include Visualbert,[15] UNITER,[16] and Vilbert.[17] With ViT, region proposals were gradually replaced by patch-based image embeddings, as in the METER model.[18]

The second breakthrough was the introduction of contrastive language-image pre-training (CLIP),[19] which is widely regarded as the first foundation model in CV. CLIP addressed a long-standing limitation of conventional vision models, which were restricted to recognizing object classes explicitly seen during training. What set CLIP apart from preceding approaches was its use of large-scale contrastive learning, enabling open-vocabulary recognition, that is, the recognition of objects the model was not explicitly trained for. At a time when Large Language Models (LLMs) had already demonstrated strong generalization capabilities, CLIP filled a comparable role on the vision side. Trained on 400 million image-text pairs, CLIP exhibits strong zero-shot and few-shot performance and has been widely used for open-vocabulary image classification, tagging, and retrieval.

CLIP inspired a series of related VLMs, including ALIGN,[20] ALBEF,[21] SimVLM,[22] BLIP,[23] and SigLIP.[24] These models are primarily designed for the task of open vocabulary image captioning, which can translate in bridge inspections for automated classification and description of inspection-relevant objects without training on costly to prepare datasets. However, these models were trained on millions of general data from the Internet; therefore, their capabilities on special-field data, such as bridge inspections, must be validated.

Several CLIP-inspired models extended the range of supported VL tasks. GLIP,[25] for example, unified bounding-box object detection with phrase grounding and with GLIPv2[26] the grounding capability was extended to instance segmentation. CLIPSeg[27] enabled object segmentation from text and visual prompts. Together, these capabilities offer significant potential for efficient detection without costly task-specific training and for the efficient annotation of damage images.

Alayrac et al.[28] argued that VLMs based on contrastive learning are limited in both language generation and task coverage and suggested an LLM-based model. Since around 2023, many VLMs have followed a broadly similar paradigm: image and text representations, generated by Transformer-based encoders, are fused to multimodal embeddings which are then passed to decoder layers of a pretrained LLM to generate the final textual outputs. The use of pretrained LLMs represents a significant shift in VLM architectures. Fully end-to-end training of VLMs has become less common. Across different architectures, the training strategies vary in terms of which components are pretrained from scratch, fine-tuned, and kept frozen. Figure 2 illustrates the general architectural design of large VLMs with example inputs and outputs.

Through the integration of LLMs, reasoning capabilities and general knowledge in VLMs enhanced such that the understanding of complex scenes has increased. Prominent recent models following the paradigm of LLM usage in VLM architecture, include PaLI[29] which has several extensions, such as PaLIGemma[30] being capable of VQA for remote sensing and phrase grounding with object segmentation. Other models that support general visual grounding with bounding boxes or segmentation masks include KOSMOS-2,[31] CogVLM,[32] GroundingDino,[33] GLaMM,[34] and SAM 3.[35] Further notable VLMs are LLaVa[36] and Qwen,[37] and their successors, respectively. Models, such as SmolVLM[38] and StreamingVLM,[39] extend image understanding to video understanding. Video understanding is particularly relevant for bridge inspections conducted via unmanned aerial vehicles (UAVs), as these typically involve long video sequences and require object detection and interpretation across multiple views and scales.

Many of these VLMs offer strong zero-shot and few-shot capabilities and can be adapted to diverse tasks including image captioning, VQA, and visual dialog, which are important for different tasks in bridge inspections such as detection and localization of damage, inspection report generation, and inspector-AI collaboration for targeted information retrieval.

3 | METHODOLOGY

This review was conducted following the PRISMA guidelines for systematic literature reviews. The study selection process is summarized in the PRISMA flow diagram shown in Figure 3. A literature search was conducted in July 2025 using the Scopus and Web of Science databases. The following Boolean search query was used:

```
("vision language" OR "vision-language" OR VLM OR
"multimodal model*"
 OR MLLM OR VLP OR CLIP OR "image caption*"
 OR "visual question answering" OR VQA
 OR "visual grounding" OR "visual dialog*")
AND
 (bridge* OR "bridge inspection*" OR "structural
inspection*"
 OR "infrastructure inspection*" OR inspect*)
AND
```

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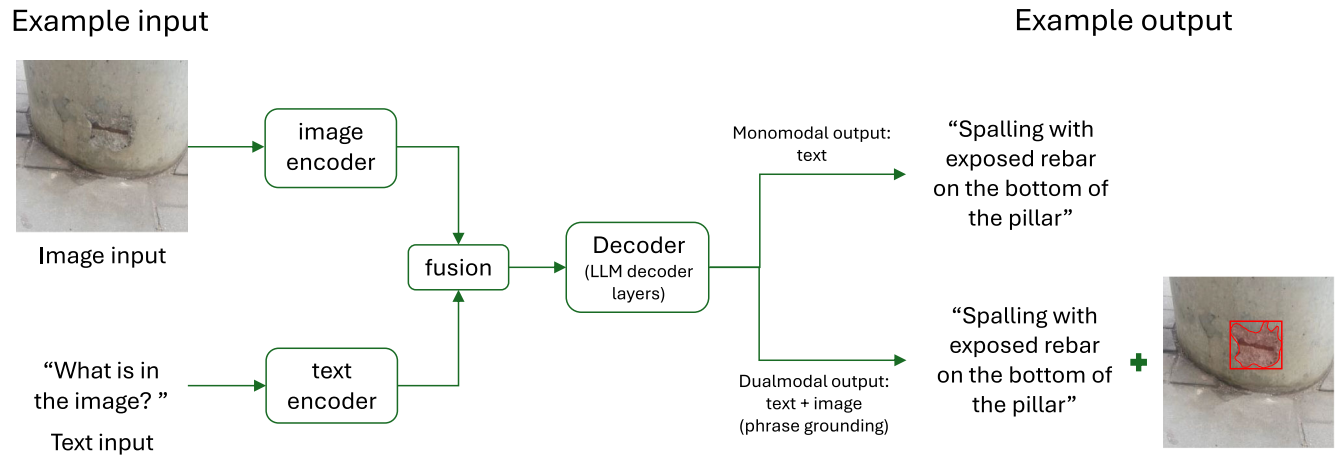


FIGURE 2 General architectural design of VLMs with example input and output. Two output options are possible: (1) Monomodal, that is, only text output and (2) dual-modal output, that is, text and image output with detection marks (bounding box or segmentation masks). To date, only a subset of models is capable of dual-modal output.

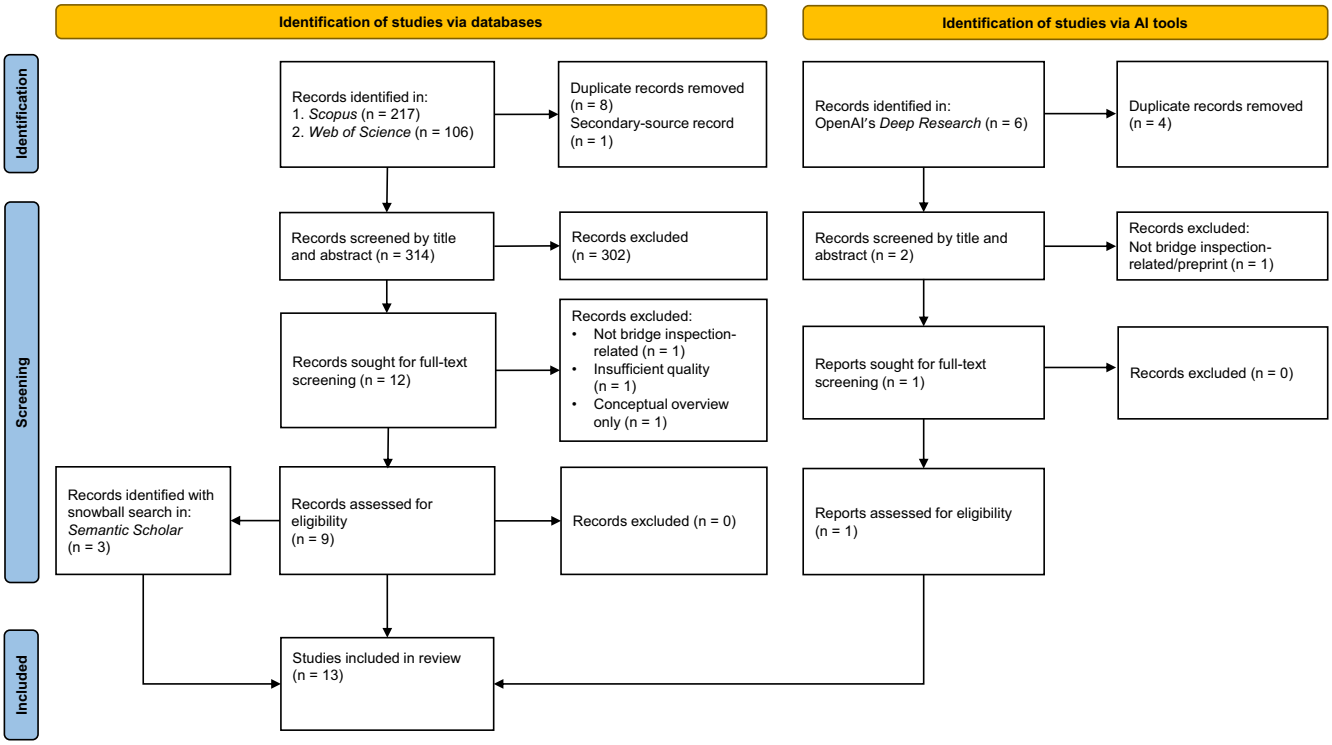


FIGURE 3 PRISMA flow diagram of the systematic literature search and study selection process for VLM applications in bridge inspection and damage documentation. The diagram includes all searches in Scopus, Web of Science, Semantic Scholar, and OpenAI's Deep Research, adapted from.[40]

```
(damage* OR defect* OR crack* OR deteriorat* OR spall*
OR corrosion OR "surface distress" OR "condition
assessment*"
```

The expanded forms of the abbreviations used in the search query, along with other abbreviations used in this work, are provided in the Online Appendix S1 [Correction added on 30 July 2026, after first online publication: The preceding sentence has been modified to remove the citation to Appendix S2.]. Search results were restricted to: (1) Subject areas: Engineering and Computer Science, (2) Document types: journal articles, conference papers, and reviews, and (3) Language: English. No time restrictions were imposed due to the recent emergence of VLMs.

The Scopus search returned 217 records. After title and abstract screening according to the inclusion criteria, 12 records remained. The Web of Science search yielded 106 records, of which nine remained after title and

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abstract screening. Eight of these were duplicates already identified in Scopus. The remaining record, a review article by Pan et al.,[3] was excluded as it represents a secondary source rather than original research.

Full-text screening was conducted for the 12 unique studies. Three papers were excluded because (1) they focused on building inspection, (2) lacked sufficient methodological detail or scientific quality, and (3) provided a conceptual overview without original contributions. This resulted in nine primary studies.

To further improve coverage, a forward-backward snowball search was performed using Semantic Scholar by examining reference lists and citing documents of the included studies. This procedure identified three additional eligible studies, yielding a total of 12 papers.

Additionally, OpenAI's Deep Research system was employed as a supplementary search tool to mitigate the risk of missing studies. The following prompt was used: "Search for literature in the field of vision-language model application to damage detection and bridge inspections. Look only for academic papers. The bridge type and damage type do not matter. They can be specified or not." Clarifying instructions specified that (1) no publication date range should be prioritized, (2) top-tier AI conferences need not be weighted more heavily than other sources and arXiv preprints could be included, and (3) pretrained VLMs such as CLIP and BLIP should be included. Deep Research identified six studies. Four were duplicates already captured from the previous database searches. All papers listed by Deep Research were manually verified via the provided links and Scopus thereby removing the risk of hallucinated references. Of the two non-duplicate papers, one was excluded as it concerned road damage and was available only as an arXiv preprint. The remaining study was peer-reviewed and not captured by prior search steps and was therefore included.

In total, 13 studies met all inclusion criteria and were included in the review. A repeat search conducted in Scopus and Web of Science in November 2025, with the identical screening procedures, yielded no additional studies, indicating full coverage of the relevant literature.

4 | LITERATURE REVIEW OF VLM APPLICATIONS IN BRIDGE INSPECTION

The application of VLMs to bridge inspection is a recent development, with the first study appearing in 2022. Since then, research activity has increased slowly but diversified into four VL task categories: (1) descriptive tasks (image captioning), (2) recognition tasks (classification), (3) query-based tasks (VQA), and (4) interpretive tasks.

Figure 4 presents the temporal distribution of the 13 identified studies across these categories. The field saw only isolated activity in 2022 and 2023, followed by a marked rise in 2024, where seven studies appeared across all four task types. Interest has since diminished, with four publications up to November 2025. This pattern suggests not only an exploratory phase but also a relatively slow adaptation of VLMs to bridge-inspection research, despite the breadth of capabilities these models offer. Geographically, the field is concentrated in East Asia: Japan accounts for five studies, followed by China (three), the United States (two), and Hong Kong, New Zealand, and the United Kingdom (one each).

The following subsections review the 13 studies according to the four identified VL task categories. This categorization provides a structured lens through which to understand how VLM capabilities have been adapted for bridge inspection and enables comparison of motivations, methods, and outcomes across the literature.

4.1 | Descriptive tasks: Image captioning for damage documentation

Image captioning represents the earliest and still one of the most applied VLM applications in bridge inspection. The underlying motivation across studies is consistent: bridge damages often require textual description rather than mere class labels, making captioning a natural extension of traditional detection workflows.

The first contribution in this space was made by Chun et al.,[41] who developed a custom VLM combining a CNN for visual feature extraction with a gated recurrent unit-based decoder to generate explanatory captions. Trained on a large dataset of Japanese inspection reports (MLIT dataset), the model produces short descriptions

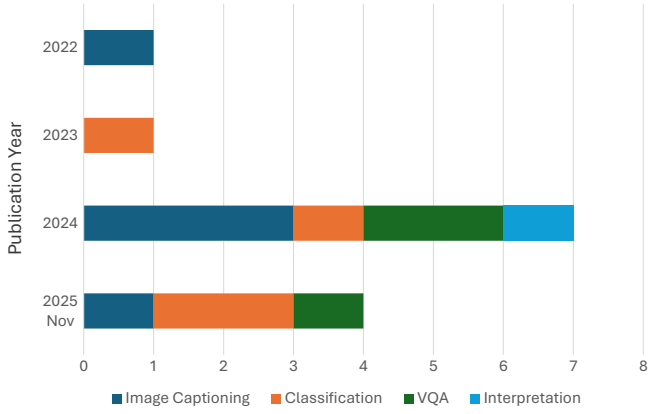


FIGURE 4 Trend in publications on VLM-based damage documentation until November 2025 categorized into four VL tasks: Image captioning, classification, VQA, and interpretation.

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indicating the damage type and corresponding bridge member. Their follow-up work[42] integrated a similar image captioning model into a tablet-based web platform, demonstrating an early attempt to embed VLM-derived outputs directly into an inspection user interface. The study primarily emphasized the technical implementation of the platform and the iterative improvement of the image captioning model through data collection via the interface. As a result, aspects related to user experience, including qualitative feedback from inspectors and quantitative usability metrics, were not explored.

Building on this foundation, Li et al.[43] identified two key shortcomings in Chun et al.'s approach. First, the model generated only short phrases rather than full, semantically rich sentences. Second, it lacked spatial localization of damage within the image. To address these issues, the authors introduced BDCD-Net, a custom encoder-decoder architecture that produces longer, more descriptive captions and enables phrase grounding by incorporating attention maps to localize relevant regions. Their model outputs information regarding the damage category, the affected bridge member, the relative location of the damage, and the camera's shooting angle. The authors themselves note that BDCD-Net cannot estimate damage severity or produce damage ratings. However, this information typically requires geometric measurements that cannot be obtained from RGB images alone. Furthermore, the purpose and operational value of reporting the camera angle remain unclear.

A more comprehensive attempt to address limitations of captioning was proposed by Chai et al.,[44] who introduced the Domain Knowledge-Driven Image Captioning Framework (DKIC-Net). Their work is motivated by the problem of hallucination: the generation of plausible but incorrect details in image captions. They attribute hallucinations to three main factors: inadequate feature representation, limitations in attentional mechanisms that fail to model interdependencies in image regions, and the lack of domain-specific knowledge in pretrained weights derived from general-purpose datasets. DKIC-Net introduces a dual-branch network combining Faster R-CNN with a CNN-based architecture and employs a dynamic attention mechanism designed to capture damage correlations better. Domain knowledge is incorporated by initializing Faster R-CNN with weights from a prior damage-detection training. Although the architecture is reported to significantly reduce hallucinations and to outperform prior captioning models proposed by Chun et al.[41] and Li et al.,[43] it remains unclear whether the model can also perform phrase grounding via bounding boxes, as this functionality is implied but not explicitly shown.

In parallel to these custom architectures, Wang and El-Gohary[45] explored a different direction by training the foundation model BLIP to generate free-form textual descriptions. Their motivation stems from limitations in existing detection approaches: (1) the heavy reliance on manual expert annotations, (2) the lack of explicit modeling of relationships between bridge components and associated damages, and (3) the constraints imposed by predefined class labels. They argue that the most informative output for inspectors is a descriptive sentence that articulates the relationship between damage and component. To this end, they fine-tuned BLIP on 3000 image-text pairs extracted from inspection reports. While the model generated mostly partially correct descriptions, it also produced missing information, spatial inconsistencies, and over-elaborated content.

These architectural and task-design differences are directly reflected in how model performance is evaluated and reported across studies. Table 1 summarizes the reported performance of the reviewed image-captioning models. Across the reviewed studies, training dataset sizes span more than three orders of magnitude, ranging from a few hundred to over 400,000 image-text pairs, reflecting substantial differences in data availability and annotation practices. Most studies report BLEU-1 to BLEU-4 scores[46] as primary evaluation metrics, with values typically ranging between approximately 0.68 and 0.86.

However, several captioning approaches follow a fixed description pattern, commonly structured as [damage type]-preposition-[bridge component]. Under such constrained settings, BLEU scores are well suited and tend to be high, as the space of valid textual realizations is narrow. In contrast, Wang and El-Gohary[45] target free-form textual descriptions and therefore refrain from using BLEU-based evaluation, arguing that token-level overlap metrics inadequately capture semantic correctness when multiple synonymous or structurally diverse descriptions are valid. To address this, they report semantic similarity metrics, namely CIDEr[47] (171.8) and SPICE[48] (25.5), which emphasize content-level rather than syntactic overlap.

A rare opportunity for cross-study comparison is provided by Chai et al.,[44] who reimplemented the architectures proposed by Chun et al.[41] and Li et al.[43] and evaluated all models on the common BiCap dataset. Under similar evaluation conditions, all three models achieve comparable BLEU performance, with the proposed DKIC-Net consistently outperforming the replicated model results. This result suggests that, once the evaluation dataset is identical, architectural refinements yield incremental gains, and that much of the apparent performance variation reported in the literature could be attributed to differences in datasets and evaluation protocol rather than to fundamentally superior model designs.

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TABLE 1 Reported performance of image-captioning models for bridge damage documentation.

Study and Model	Dataset: Size and scope	Metrics	Scores
Chun et al.[41] –"Explanatory text generation model"	MLIT dataset: 422,290 img-txt pairs, 27 dmg clss, 38 component cls	BLEU-1—BLEU-4	0.782, 0.749, 0.711, 0.693 0.831, 0.798, 0.776, 0.754[a]
Chun et al.[42] –"Explanatory text generation model"	MLIT dataset	BLEU-1—BLEU-4	0.85–0.70[b]
Li et al.[43] –BDCD-Net	2876 img-txt pairs: 15 dmg cls, 13 component cls	BLEU-1—BLEU-4	0.842, 0.785, 0.737, 0.694 0.842, 0.813, 0.790, 0.761[a]
Chai et al.[44] –DKIC-Net	BiCap dataset: 391 img-txt pairs, 6 dmg cls, 25 component cls	BLEU-1—BLEU-4	0.868, 0.826, 0.798, 0.776
Wang and El-Gohary[45] –BLIP	3000 img-txt pairs, free-form descriptions	CIDEr, SPICE	171.8, 25.5

[a]Results reproduced on the BiCap dataset and reported in Chai et al.[44]
[b]Exact values were not provided; BLEU performances were reported in a plot.

While these gains are incremental, their practical relevance for large-scale inspection workflows has not yet been systematically evaluated.

Altogether, these studies demonstrate the potential of image captioning for bridge inspections, while also exposing consistent and unresolved limitations. Generated captions frequently remain incomplete, hallucinations persist across architectures, visual grounding is rarely integrated, and none of the reviewed models provide information on damage extent, despite its central importance for subsequent documentation and assessment tasks. From a performance perspective, reported quantitative results remain difficult to compare due to differences in datasets, output constraints, and evaluation protocols. Taken together, these limitations underscore the need for fine-tuned VL foundation models capable of free-form and inspection-relevant description, closer alignment between captioning outputs and operational information requirements, and, critically, the establishment of a bridge inspection benchmark dataset to enable meaningful performance evaluation.

4.2 | Recognition tasks: Damage and component classification

Several studies explored VLMs not for descriptive generation but for recognition tasks, reducing the problem of bridge-damage capture to classification. Although classification provides a simplified entry point for VLM integration, it does not exploit the full potential of VL models and limits the output to predefined categories.

Ma et al.[49] introduced a model termed Finetuned CLIP (FTC), which combines a CLIP-based image encoder with the visual backbone of a single-stage object detector and augments the fused embeddings with a classification layer. Their model was trained to detect bounding boxes and classify four damage types, that is, cracks, spalling, scaling, and efflorescence, and achieved higher accuracy and recall than the comparison model YOLOv8. The authors argue that integrating CLIP improves robustness against adversarial variations; however, FTC is reported to struggle with ambiguous or visually uncertain damage cases. More fundamentally, classification is not a task that fulfills the information requirements of many damage categories, and reliance on predefined labels prevents leveraging CLIP's open-vocabulary capabilities.

In contrast to modifying CLIP's image-encoder, Liao and Nakano[50] focused on the text encoder of CLIP. They proposed BridgeCLIP, which introduces a Description Attention Module (DAM) based on two parallel CLIP text encoders, one processing natural-language damage descriptions and the other processing learnable prompts. The following two representations are then fused through an attention module, which is learnable as opposed to the frozen text and image encoders of CLIP. Their motivation lies in the challenge of learning relationships among different damage types, though it is not clearly explained which relationship is meant and reducing reliance on large, annotated datasets. BridgeCLIP is reported to yield performance gains over standard CLIP while avoiding heavy fine-tuning. Nevertheless, like FTC, the model is only trained and tested on a small set of predefined classes.

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TABLE 2 Reported classification performance of VLM-based approaches in bridge inspection.

Study and Model	Dataset: Size and scope	Acc.	Pr.	Re.	F1	Notes
Ma et al.[49] –FTC	12,473 imgs, 5 dmg cls	—	—	—	—	CLIP-based classification. Per-class accuracy: 3 cls >99%, one cls 94%, one cls 81%
Liao and Nakano[50] –BridgeCLIP	11,510 imgs, 13 dmg cls	—	51.75	62.33	55.94	CLIP-based classification
Chen et al.[51] –Few-shot CLIP with prompt learning and cache adapter	924 imgs, 8 component cls + 4257 imgs, 2 dmg cls	83.33	—	—	—	UAV-oriented component and damage recognition
Wang et al.[53] –Claude	1798 imgs, 2 dmg cls	93.5	96.2	85.7	89.7	Prompted MLLM
Wang et al.[53] –GPT-4o		85.4	82.1	83.4	82.7	Prompted MLLM
Wang et al.[53] –GPT-4o mini		89.2	87.2	84.7	85.8	Prompted MLLM
Wang et al.[53] –Grok		84.9	85.4	83.7	84.2	Prompted MLLM
Wang et al.[53] –Gemini		83.8	83.5	83.8	83.6	Prompted MLLM

Note: Accuracy (Acc.), Precision (Pr.), Recall (Re.), and F1-score are reported in percent.

A more operationally oriented perspective was adopted by Chen et al.,[51] who examined CLIP regarding a UAV-based inspection workflow. Their objective was to enable UAVs to interpret and respond to the inspection environment through human-issued text prompts. Using Context Optimization (CoOp),[52] a prompt-learning method, they fine-tuned CLIP on datasets of bridge components and damage types. While their approach demonstrates promising adaptability via few-shot learning, two limitations are noteworthy: (1) component images were preprocessed by masking out the background, simplifying the visual content, which diverges from real UAV imagery, and (2) the proposed UAV-VLM system was not validated in real UAV inspection flights, leaving its practical viability untested. The authors themselves note the lack of interpretability in CoOp-generated prompts and suggest knowledge-graph-based alternatives.

Beyond CLIP-based approaches, Wang et al.[53] evaluated the crack detection performance of commercial multimodal large language models (MLLMs) for crack detection. In a two-stage pipeline, they first applied image-level classification to detect crack-containing images, followed by patch-wise classification of images to obtain coarse segmentation. They benchmarked five MLLMs, including Claude, GPT-4o, GPT-4o mini, Grok, and Gemini, against five supervised CV models, including Segmented Transformer, Swin Transformer, ResNet, EfficientNet, and MobileNet. While the MLLMs performed comparably to supervised models at the image level, their performance deteriorated substantially when evaluated at the patch level. They highlight the advantages of MLLMs, particularly the elimination of dataset annotation and model training, yet acknowledge that crack-specific fine-tuning is still required for practical deployment. A critical omission, however, is the lack of discussion on inference cost and hardware constraints, which are key factors for mobile and UAV-based inspection.

This work by Wang et al.[53] recalls earlier efforts in coarse segmentation via patch-wise classification, such as the sliding-window approach employed by Cha et al.[54] in the late 2010s. However, it is now widely recognized that accurate segmentation is essential for crack localization and quantification. In this context, evaluating the models on crack segmentation would have been more appropriate and represents a key direction for future research.

Table 2 summarizes the reported classification performance of the studies applied to damage recognition and, in some cases, bridge component identification. Dataset sizes span more than two orders of magnitude, ranging from a few hundred to over 12,000 images, and differ substantially in terms of annotation scope, class definitions, and preprocessing. These differences strongly influence the reported performance values.

Two of the reviewed studies report accuracy as their sole evaluation metric. Accuracy is generally ill-suited for imbalanced classification problems, which all reviewed datasets exhibit to varying degrees. Notably, studies

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relying on accuracy report comparatively high values exceeding 83%. These results coincide with experimental setups that simplify the visual content, for example by masking backgrounds or tightly cropping objects of interest.

In the study by Chen et al.,[51] performance is reported without class-wise breakdowns, limiting insight into how reliably different component or damage classes are recognized. In addition, the evaluation does not explicitly address the multi-scale concept of the proposed framework, in which images range from component-level to close-up damage views acquired at different inspection time steps. Consequently, the robustness of the proposed approach under realistic image sequences in UAV-based inspections remains unclear.

By contrast, Liao and Nakano[50] report standard classification metrics, including precision, recall, and F1-score. For 13 damage classes, the reported values (51.75%, 62.33%, and 55.94%, respectively) are substantially lower than those observed in crack-specific studies, reflecting the increased complexity of multi-class damage recognition. Wang et al.,[53] focusing exclusively on binary crack detection, report F1-scores above 82% for multiple MLLMs. However, the reported F1-scores of the supervised CNN- and ViT-based classifiers achieve higher performance while being considerably smaller and less computationally intensive.

Overall, the reported performance results underline the strong dependence of classification performance on task formulation, data preprocessing, and number of classes. High performance is primarily observed for narrowly defined recognition tasks under controlled conditions and simplifying assumptions, whereas performance degrades substantially for more realistic multi-class scenarios. Across the reviewed literature, VLM-based classification approaches underutilize the open-vocabulary capabilities of VLMs, which can lead to a disproportionate cost–benefit ratio given their substantial computational overhead. Notably, this overhead is rarely reported or systematically analyzed.

4.3 | Query-based tasks: Visual question answering (VQA)

VQA represents another important direction in the automation of inspection processes, particularly for tasks requiring targeted information retrieval rather than free-form description. Three of the reviewed studies applied VQA, each motivated by a distinct operational need.

Yamane et al.[55] used VQA to infer potential causes of damage directly from images, arguing that causal assessment often relies on contextual information beyond the immediate damaged region. For example, corrosion caused by leaking drainage pipes cannot be identified if the pipe is not visible in the image. To address the issue of image scale and missing context, the authors combined a VQA model (MCAN[56]) with Structure from Motion (SfM) under a UAV-based inspection scenario. Their pipeline first applies VQA to the target image to determine the damage type and affected bridge member. Based on this information, the system retrieves a predefined set of possible causes. SfM is then used to locate surrounding images, on which VQA is run again to check whether relevant members and causal indicators are present. The approach was demonstrated using UAV-acquired images of steel bridges for corrosion assessment. While the authors briefly mention limitations, such as dynamic camera movement and dependence on predefined cause definitions, they do not provide a detailed discussion of failure cases. A more thorough characterization of causal inference challenges, including the broader domain knowledge required to support such tasks, would have made the classification of this study easier.

Kunlamai et al.[57] also explored VQA for bridge inspection but from a different standpoint. They argued that image captioning often produces generic descriptions that may not match the information needs of inspectors, whereas VQA enables focused querying. Their work investigated the effectiveness of a two-step pretraining-fine-tuning strategy on improving VQA performance. Using established VQA architectures, including ALBEF and three variants of MMBERT,[58] they first pretrained the models on a subset of image-text pairs from the MLIT dataset[41] using standard VL objectives: masked language modeling, image-text matching, and contrastive learning. They then fine-tuned the models on a smaller dataset of question-answer pairs derived from inspection images. This two-step training improved performance while reducing dependence on costly manual QA annotations, a challenge also highlighted previously by Yamane et al.[55] However, several difficulties arose. Class imbalance led to misidentification of bridge members, ambiguous ground truth definitions caused confusion (e.g., anchor bolts vs. bearings), and issues of image scale persisted, as many close-up images lacked the context required for correct responses. Although the study demonstrates the feasibility of VQA for targeted information extraction, the relatively simple question types, regarding the bridge type and bridge member, raise the question of whether such tasks could be handled more efficiently through captioning rather than VQA.

Zhang and Liu[59] proposed the three-stage framework, DamageQwen, for pavement-damage documentation. Their pipeline begins with YOLO v11x to detect cracks.

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False positive bounding boxes are filtered using CLIP. The resulting bounding-box coordinates and crack categories are then passed to the off-the-shelf VLM Qwen2-VL using structured prompts, enabling the generation of structured documentation. This approach effectively provides the model with artificial phrase grounding, compensating for the model's inability to localize, detect, or quantify cracks using raw images alone. Few-shot learning on a small, annotated dataset was further used to improve documentation accuracy. While the authors note that the model can provide answers at varying levels of detail, the operational scenarios requiring such granularity are not clearly articulated. Moreover, severity assessments remain qualitative, relying on pixel-based bounding-box coordinates rather than real-world measurements, and the multi-stage structure introduces a risk of upstream error propagation.

Table 3 summarizes the reported performances in VQA studies. A direct quantitative comparison is limited as the evaluation setups differ fundamentally in terms of task formulation, dataset scale, and output structure.

The studies by Kunlamai et al.[57] and Yamane et al.[55] evaluate VQA primarily as a closed-set answer prediction task on the large-scale MLIT dataset. Accordingly, performance is reported using overall accuracy, with values ranging from approximately 78.9% to 88.3% across different model architectures. It remains challenging to achieve near-perfect consistency in answers, particularly when questions do not match the spatial scale of the image. In contrast, Zhang and Liu[59] frame VQA as a free-form language generation problem and evaluate their approach using multiple text-similarity metrics, including ROUGE, BLEU, METEOR, and Bert-Score. Their study is conducted in a few-shot setting, with only four image-text pairs used for fine-tuning. While the high Bert-Score suggests strong semantic overlap between generated and reference answers, substantially lower ROUGE and BLEU values suggest considerable variation in wording and phrasing. These metrics raise an open research question as to whether semantic similarity metrics adequately capture inspection-and-infrastructure-specific vocabulary and to what extent linguistic variability is compatible with the highly specific terminology required in formal bridge inspection documentation.

Collectively, the reviewed VQA studies reveal substantial limitations. These include persistent challenges with image scale, reliance on predefined question templates, and limited handling of complex causal reasoning. In addition, the practical value of VQA is not always clearly justified: in two of the studies, the underlying use cases are not well defined, with tasks that might have been addressed more appropriately through captioning or structured text generation rather than question answering. As such, the meaningful application of VQA in operational bridge inspections requires well-defined use cases. In particular, the use of large models must be proportionate to the intended application.

4.4 | Interpretive tasks: Multimodal reasoning for interpretation

A final category of VLM applications focuses on interpretive tasks, where models are expected to move toward higher-level reasoning. In this context, Darji et al.[60] examined how MLLMs and VLMs perform when

TABLE 3 Reported VQA performance of VLM-based approaches for bridge inspections.

Study and model	Dataset: Size and scope	Metrics	Score
Yamane et al.[55] –Deep modular co-attention network (MCAN)	MLIT dataset +20,079 UAV imgs	Accuracy	81.2
Kunlamai et al.[57] –Baseline VQA (ResNet152)	MLIT dataset	Accuracy	78.85
Kunlamai et al.[57] –MMBERT (ResNet152)		Accuracy	79.01
Kunlamai et al.[57] –MMBERT (CLIP-RN50)		Accuracy	82.09
Kunlamai et al.[57] –MMBERT (CLIP-ViT-B/16)		Accuracy	85.41
Kunlamai et al.[57] –ALBEF (ViT-B/16)		Accuracy	88.30
Zhang and Liu[59] –DamageQwen	4 img-txt pairs (few-shot learning)	ROUGE-1, ROUGE-2, ROUGE-L, BLEU, METEOR, Bert-Score	0.4828, 0.2162, 0.3701, 0.1494, 0.3494, 0.9099

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interpreting subsurface inspection data derived from non-destructive evaluation (NDE) techniques. This work differs from the rest of the reviewed literature in that the visual input is not inspection imagery but specialized technical imagery in the form of contour maps generated by ground penetrating radar (GPR), electrical resistivity (ER), impact echo (IE), and ultrasonic surface waves (USW).

The authors evaluated two task types: (1) image captioning, where the model generates simple descriptions of the contour maps, and (2) summarization, where the model synthesizes insights into bridge condition and recommended actions.

For captioning, a broad set of models was assessed, including Claude 3.5 Sonnet, ChatGPT-4, CogVLM2, ShareGPT4V, Florence-2, Paligemma (standard and fine-tuned versions), BLIP large, and vit-gpt2. Models such as ShareGPT4V, CogVLM2, ChatGPT-4, and Claude 3.5 demonstrated comparatively strong performance and were therefore selected for the more demanding summarization task. In this task, ChatGPT-4 achieved the highest overall rating (5.0/5), followed by Claude 3.5 (4.67/5), with both models producing technically nuanced, though not consistently reliable, interpretations of the contour maps.

The performance evaluation relied on a custom, expert-driven scoring scheme. Summaries were assessed according to three criteria, that is, completeness, depth of coverage, and formatting/presentation, each rated on a 5-point scale. However, the evaluation procedure is only partially specified. The performance was judged by researchers and an NDE technician. As a result, the evaluation is difficult to reproduce or scale.

Despite these encouraging results, the study has three limitations. First, prompt design and interaction procedures are not described in sufficient detail. Although prompt optimization is mentioned as a central component of the study, only a single full prompt is provided, and it remains unclear whether session resets and user settings were used when responses were unsatisfactory. Second, the dataset is small, comprising only five contour maps, limiting the generalizability of the findings and preventing deeper analysis of model consistency or failure patterns. A more extensive evaluation could have yielded stronger insight into how MLLMs interpret technical NDE imagery. Third, the reliance on expert-based manual evaluation limits scalability, as the assessment cannot be partially automated or applied to larger datasets.

5 | DISCUSSION

The reviewed studies reveal a consistent misalignment between the potential of VLMs and their current application in bridge inspection research, which are examined in the following subsections: (5.1) characteristics and limitations of training data, (5.2) information requirements for automated damage documentation, (5.3) structural context and multi-modal evidence in damage documentation, and (5.4) VLM implementation strategies and architectural implications.

5.1 | Training data for VLM application

A review of the datasets used across the selected studies reveals substantial variation in data origin, scale, annotation quality, and label coverage (Table A1 in Appendix A). Among the 13 studies, nine relied on datasets derived from inspection reports. Four of these[41,42,55,57] used the large-scale dataset provided by Japan's Ministry of Land, Infrastructure, Transport and Tourism (MLIT), comprising 422,290 image-text pairs spanning 27 damage types and 38 component classes. This dataset remains the most comprehensive resource available and has shaped much of the existing research, particularly in the work of Chun and Yamane, who appear as coauthors on all studies drawing on MLIT data.

Other inspection-report-based datasets are markedly smaller, typically 2800–7400 image-text pairs, with fewer damage and component categories.[43,44] Component classes tend to be more richly represented than damage types, likely reflecting the diversity of structural materials in the source bridges (e.g., steel, reinforced concrete, prestressed concrete). While the MLIT dataset specifies bridge-type composition, this information is generally missing in other reported datasets.

Compared with datasets typically used for image detection tasks, datasets based on inspection reports offer two important advantages: larger size and broader class coverage. Their accompanying textual descriptions also serve as natural annotations and reduce the need for manual labeling. However, these datasets also entail challenges that are often overlooked. Inspection texts often contain quantitative and contextual information, such as dimensions, quantity of damage, or bridge component numbering, that is not directly reflected in the images. Only Wang and El-Gohary[45] and Chai et al.[44] explicitly describe preprocessing procedures to align textual metadata with corresponding visual content.

Beyond inspection-report-based datasets, four studies[50,51,53,59] used publicly available datasets. These datasets typically focus on a single damage type (e.g., cracks or corrosion), lack component annotations, and include hundreds to a few thousand images. Despite the availability of segmentation masks or bounding-box annotations, the reviewed studies generally did not leverage

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these annotations. Instead, annotations were often simplified to text descriptions for classification purposes. Two studies[51,55] augmented public or inspection-report data with self-collected images to expand category coverage or adapt to UAV-based settings.

Although inspection archives and BMS repositories contain extensive image-text pairs that could, in principle, support VLM training, some structural limitations may restrict their immediate usability: these datasets were not created with ML workflows in mind. They reflect human inspection practices, in which images and texts are complementary rather than redundant. This often leads to dataset imbalance, variable image quality (e.g., due to compression or resizing), and, most critically, persistent issues with image scale. Image scale refers to the extent of contextual information surrounding a damage: if too little context is captured, the affected component may be unidentifiable. If too much context is present, unrelated surrounding features can lead to misclassification. As a result, while such datasets provide a valuable foundation, they are insufficient as the sole, unprocessed data source for inspection automation with VLMs.

Finally, the dominance of inspection datasets from East Asia[41,42,55,57] and a small number of US studies[45,60] highlights the geographical concentration of current VLM research. Countries with well-established BMSs are underrepresented, and the field lacks open, country-specific benchmark datasets aligned with local documentation standards and languages. Such datasets are essential not only for model training but also for standardized evaluation across regions. Achieving this, however, also presupposes clearly defined information requirements for describing and documenting damage, which remain largely unspecified in current research.

5.2 | Information requirements for automated damage documentation

A fundamental limitation across current studies is the absence of clearly defined damage-information requirements, that is, a shared understanding of what constitutes a complete and operationally useful inspection record. Without such standardized requirements, it is difficult to evaluate, compare, or meaningfully position existing VLM frameworks within actual inspection workflows.

The reviewed studies predominantly focus on identifying the damage type and the affected bridge component, thereby addressing only a small subset of the information needed for comprehensive documentation. In practice, an operational inspection documentation optimally encompasses at least the following elements:

1. Damage type and textual description
2. Component type and spatial location
3. Damage quantification, i.e., size (e.g., crack width and length) and amount
4. Severity or condition rating
5. Temporal damage development based on historical records
6. Damage cause
7. Forecast of likely damage development
8. Recommended repair or maintenance action

From a broader asset-management perspective, the rehabilitation plan may be considered as a ninth element. For the eighth and ninth elements, cost estimations are an important aspect and of high relevance for bridge owners. This set of elements serves as an analytical reference framework for reviewing existing research and pinning down studies to come.

Table 4 provides an overview of how the reviewed studies cover the different information elements. Across the 13 studies, all address element (1), that is, damage type, with three studies focusing exclusively on this element. Nine studies address damage type in conjunction with element (2), component type. Only three studies consider additional elements beyond damage and component type, including damage cause (6), repair recommendation (8), damage quantification (3), and severity rating (4). The study covering the largest number of elements is Zhang & Liu,[59] which addresses five elements; however, its scope is limited to pavements. More generally, the covered elements are typically addressed in a narrow manner, as discussed in Section 4. In particular, damage type is often treated in a simplified way, while complex real-world damage scenarios requiring higher-level interpretation and summarization remain largely unaddressed, as illustrated in Figure 1. These aspects have not yet been systematically considered in the reviewed literature.

For certain damage types, most notably cracks and spalls, textual description alone is insufficient, as geometric annotation remains essential for quantification. While VLMs equipped with visual grounding capabilities can link textual descriptions to image regions, this represents only an initial step. Visual grounding enables the localization of damage within the image, but determining its size requires metric reasoning and additional modalities. This may involve depth estimation or the interpretation of measurement tools that serve as reference objects, such as folding rulers or crack gauges, along with their associated measurement values. Notably, Wang et al.[53] provided a small-scale study about whether VLMs, including GPT-4o and Claude, can read from folding rulers and handwritten measurement annotations on bridges. However, none of the reviewed studies systematically

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Study	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Chun et al.[41]	✓	✓	–	–	–	–	–	–
Li et al.[43]	✓	✓	–	–	–	–	–	–
Chai et al.[44]	✓	✓	–	–	–	–	–	–
Wang and El-Gohary[45]	✓	✓	–	–	–	–	–	–
Chun et al.[42]	✓	✓	–	–	–	–	–	–
Ma et al.[49]	✓	–	–	–	–	–	–	–
Liao and Nakano[50]	✓	–	–	–	–	–	–	–
Chen et al.[51]	✓	✓	–	–	–	–	–	–
Yamane et al.[55]	✓	✓	–	–	–	✓	–	–
Kunlamai et al.[57]	✓	✓	–	–	–	–	–	–
Zhang and Liu[59]	✓	✓	✓	✓	–	–	–	✓
Wang et al.[53]	✓	–	–	–	–	–	–	–
Darji et al.[60]	✓	–	–	–	–	–	–	✓

TABLE 4 Overview of the extent to which reviewed VLM-based studies address inspection information requirements.

Note: (1)–(8) correspond to the inspection information categories defined before "(✓)" indicates that the requirement is at least partially addressed; "–" indicates that it is not addressed.

evaluated whether VLMs can recognize or interpret such measurement cues, especially in the absence of handwritten annotations. Moreover, reliable extraction of all required measurement information often necessitates multiple images. Although many large VLMs support multi-image inputs, how effectively such evidence can be integrated into a coherent and unified damage entry remains an open question.

It is also important to recognize that bridge inspection is not exclusively visual, even though it is predominantly performed through sight. Inspectors rely on multiple human senses to form a complete assessment, allowing them to perceive information inaccessible to any ML model. Table 5 summarizes these sensory modalities and their typical use cases in inspection practice.

This table highlights a crucial point: several information elements listed above depend on perceptual cues beyond vision. Consequently, VLMs, and even current MLLMs, cannot capture all sensory modalities involved in inspection practice without human assistance. While certain sensory functions, such as auditory cues, can be partially approximated by hardware (e.g., microphones for hammer-sounding), others such as olfactory, gustatory, and tactile perception remain difficult or economically impractical to replicate. This underscores that technical systems must be designed to integrate human perceptual input within ML-supported inspection workflows. Accordingly, VLM-based systems should be understood not as replacements for human inspection but as components of a multimodal, human-AI collaborative workflow, presenting a practically grounded pathway toward efficient bridge inspection processes.

Taken together, these considerations show that the information requirements for automated damage documentation extend far beyond what current VLM applications address. Yet even within the visual domain, which is foundational to bridge inspection, existing VLM approaches capture only a fraction of the required complexity. Advancing VL capabilities therefore remains essential, alongside developments in additional modalities and collaborative workflows. These information requirements form a reference point for evaluating current techniques and future directions discussed in Sections 5.3, 5.4, and 6.

TABLE 5 Human sensory modalities used in bridge inspections.

Human sense used in inspection	Modality	Use cases in bridge inspections
Sight	Visual	Identifying visible damage and defects
Hearing	Auditory	Detecting hollow or delaminated areas through hammer-sounding or tapping techniques
Smell	Olfactory	Identifying biological contamination (e.g., mold and pigeon droppings) or chemical odors
Taste	Gustatory	Legacy practice occasionally used to distinguish chloride-related efflorescence (salty) from other surface deposits
Touch	Tactile	Assessing poor surface compressive strength, dusting, or moisture by hand

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5.3 | Structural context and multimodal evidence in damage documentation

Many studies noted that existing object-detection-based models fail to capture the relationship between bridge components and associated damage. While VLMs are well positioned to address this limitation, effective damage documentation requires more than structural localization within an image. Detected damages must be positioned within the structural hierarchy of the asset, that is, according to the structural semantics defined in BMSs, linked to historical inspection records.[61] Such contextualization is essential for downstream tasks such as damage condition rating and overall structural condition assessment.[62]

Within this framework, image scale is not merely a visual characteristic but a technical parameter: it determines whether an image contains sufficient structural context for component referencing and, consequently, whether ML outputs can be meaningfully integrated into BMS records.

Structural context is also critical for causal reasoning about damage, as demonstrated by Yamane et al.,[55] who use image sequences capturing the surroundings of corrosion to infer its cause. Nevertheless, visual evidence alone remains insufficient to reliably determine underlying damage mechanisms. For many deterioration processes, causal inference requires information beyond surface appearance. Subsurface phenomena can be captured through sensor measurements (e.g., strain, vibration, or humidity) as well as through non-destructive and destructive testing ((N)DT) methods. Integrating these complementary modalities enables a better understanding of structural behavior and deterioration processes.[63]

Realizing such integration requires a common data environment, such as a building information model (BIM) or a digital twin,[64] in which visual inspection data, sensor measurements, (N)DT results, textual documentation, and historical records are spatially and semantically aligned. Embedding ML outputs within such environments could support linking and joint evaluation of all information. To date, however, none of the reviewed VLM studies address these integration challenges, indicating a substantial gap between current research practice and the requirements of operational bridge inspection.

5.4 | VLM implementation strategies and architectural implications

The reviewed studies employ three broad strategies for applying VLMs as summarized in Table 6. Most studies rely on custom architectures, reflecting task formulations based on rigid labels. A smaller number of studies apply foundation models fully or adapt them through fine-tuning. Despite the diversity of available models, the dominance of CLIP and CLIP-based architectures, from a similar development period (e.g., BLIP, ALBEF, MMBERT), indicates a comparatively slow adoption of more capable foundation models.

TABLE 6 VLM application techniques for bridge inspections.

VLM application technique	Corresponding literature
Custom VLMs based on diverse neural network architectures (CNN, GRU, LSTM, Transformer, and CLIP)	41–44,49,50,55
Trained or fine-tuned vision-language foundation models for domain-specific adaptation	45,57
Prompt engineering and tuning	51,53,59,60

These implementation strategies entail clear trade-offs. Custom architectures are typically lightweight, locally deployable, and highly specialized, yet they are sensitive to damage cases not represented in their training data and lack flexibility beyond classification and simple captioning. Foundation models, by contrast, provide a broader knowledge base and greater task flexibility and generalize better to unseen data. However, their higher computational cost often makes them unsuitable for local inference on mobile devices, which is a critical limitation given that network connectivity is frequently restricted at bridge sites. This limitation has not been addressed in any reviewed study.

Beyond implementation strategy and deployment constraints, current VLMs exhibit architectural limitations that can affect their suitability for bridge inspection tasks. Several Transformer-based crack detection studies[65–67] have shown that additional computation modules are necessary to improve the detection of fine cracks, indicating that standard Transformer architectures struggle with fine-grained damage patterns. In addition, a general VLM benchmark study[68] indicates that difficulties in translating precise spatial information into language arise at the language decoding stage rather than in the vision-encoder, which is reported to be capable of capturing details. Complementary VLM benchmark results[69] further report substantial deficiencies in spatial reasoning, that is, understanding the spatial relationship between objects, an ability relevant to inspection documentations.

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Taken together, these findings point to potential limitations for inspection-specific tasks that rely on accurate spatial perception although these aspects have not yet been systematically investigated in bridge inspection context. Advancing VLM-based bridge inspection will therefore depend on clearly defined VL tasks, and the availability of large, inspection-oriented multimodal benchmark datasets to systematically evaluate and improve VLM capabilities.

6 | TOWARD A VL BENCHMARK FOR BRIDGE DAMAGE INSPECTION

The analysis in Section 5 suggests that many limitations of current VL-based bridge inspection research stem less from model capability than from (1) ill-defined inspection tasks that do not adequately reflect real-world inspection practice and (2) the absence of inspection-aligned benchmarks. Systematic progress in the field will be enabled if inspection-aligned benchmarks are developed.

To address this gap, this section introduces a conceptual benchmark framework for evaluating inspection-relevant VL capabilities. Rather than prescribing specific datasets or model solutions, the framework defines inspection-driven task groups and corresponding design principles intended to guide the construction of benchmark datasets and enable consistent, comparable evaluation. The scope is deliberately focused on VL tasks, which constitute the primary modalities in bridge inspection practice, while the integration of other modalities is considered beyond the present scope.

6.1 | VL tasks in bridge inspections

To enable systematic evaluation of VL capabilities in bridge inspection, an ideal benchmark should evaluate a set of task-groups that reflect the information elements of inspection documentation defined in Section 5.2. These task groups are not intended to be addressed by a single VLM. Instead, they reflect capabilities that may be distributed across specialized components within inspection pipelines, including vision-only and language-only models, as well as rule-based processing steps. Accordingly, the benchmark should support the evaluation of both individual VL components and VL pipelines.

Task Group 1: Damage Description and Reporting

This group evaluates the ability to generate inspection-relevant textual descriptions from visual evidence, focusing on synthesis rather than labels:

- Generation of accurate descriptions of damages
- Interpretation of damage patterns in structural context
- Summarization of multiple co-occurring damages
- Grouping and description of recurring damage patterns across components and based on common damage cause

Task Group 2: Language-Grounded Spatial Reasoning

This group assesses the ability to relate damage to its structural and spatial context. Tasks include:

- Identification of structural components, including ancillary elements (e.g., bearings, expansion joints, barriers, etc.)
- Description of spatial relationships between damage and structural elements (e.g., "crack at bottom of girder near bearing")
- Localization of damage
- Interpretation of inspection-relevant spatial context (e.g., corrosion below a drainage outlet)

Task Group 3: Visual Grounding and Recognition

This group evaluates the alignment between language and visual geometry, which is foundational for most inspection tasks. Tasks include:

- Grounding textual descriptions in image regions (phrase grounding)
- Recognition of inspection-relevant objects and damage using bounding box detection and semantic segmentation

Task Group 4: Temporal and Cross-Inspection Reasoning

This group focuses on reasoning across inspection cycles and temporally ordered visual evidence. Tasks include:

- Identification of the same damage instance across multiple inspections
- Description and visualization of historical damage progression
- Qualitative and quantitative assessment of temporal trends and change patterns based on visual evidence

Task Group 5: Interpretive and Causal Reasoning

This group evaluates higher-level interpretation that goes beyond direct visual observation. Tasks include:

- Inference of plausible damage causes based on visible symptoms and environmental context
- Explicit recognition of cases in which causal inference is not possible from visual evidence alone, including identification of missing or insufficient information

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- Detection of anomalies, such as missing components (e.g., absent adjusting nuts) or unexpected objects (e.g., debris)
- Identification and description of potential non-compliance with codes, or deviations from the target state, based on visible evidence (e.g., improper repair)

Task Group 6: Measurement-Grounded Reasoning

This group addresses quantification tasks that support condition assessment and documentation. Tasks include:

- Identification of measurement tools or reference markers in images
- Counting of damage instances and inspection-relevant objects (e.g., number of corroded rivets)
- Extraction and interpretation of dimensional information and measurement units from visual evidence

These tasks impose specific requirements on the benchmark datasets, which must follow well-defined principles to enable both task-level and system-level evaluation.

6.2 | Design principles for inspection-oriented VL benchmark datasets

To guide the development of benchmark datasets required for VL inspection tasks and systems, four design principles are defined. These principles are derived directly from the previously defined inspection-driven task groups.

a. *Hierarchical inspection-centric organization*: Dataset instances should be organized at the inspection level rather than as isolated images, with bridges as top-level entities containing multiple inspection data over time. This reflects operational BMS structures[70] and enables component-level reasoning and cross-inspection tasks, including damage progression analysis.
b. *Multi-scale visual data*: Each damage instance should be documented through multiple viewing scales and perspectives where necessary, ranging from contextual views showing the damage within its structural environment to detailed close-ups of specific damage characteristics. This addresses the image scale problem identified in Section 4 and enables the evaluation of interpretative reasoning and spatial reasoning capabilities of VLMs.
c. *Temporal linkage across inspections*: Damage instances should maintain persistent identifiers across inspection cycles, with explicit documentation of progression. This supports the evaluation of cross-inspection reasoning tasks, such as damage re-identification, damage progress evaluation, and damage condition assessment, which are absent from current VLM studies.
d. *Annotation aligned with inspection-relevant VL tasks*: The benchmark should include diverse annotation types aligned with the task groups defined in Section 6.1, including damage definitions for prompt engineering, captions for descriptive reporting, VQA triplets for specific information extraction, and phrase grounding with image annotations for spatial localization and geometric information.

These principles define key requirements for VL benchmark datasets and have implications for their construction.

6.3 | Implications for VL benchmark dataset construction

Current VL research in bridge inspection is largely based on single-image evidence. However, many inspection-relevant reasoning tasks cannot be reliably addressed using isolated images due to inherent limitations in scale and context. Consequently, single-image datasets restrict the extent to which higher-level VL capabilities can be evaluated and, in turn, constrain progress toward higher levels of automation in inspection workflows.

Bridge inspection is inherently multi-scale and sequential. Inspectors rely on combinations of overview views, component-level views, and close-up views from direct visual observation, complemented by visual evidence from previous inspection cycles as visualized in Figure 5. A VL benchmark intended to reflect inspection practice must therefore move beyond isolated images and incorporate multi-image evidence, preferably in the form of video. UAV-based inspections naturally provide such data and are therefore particularly well-suited for VL benchmarks. While multi-image sets and videos increase annotation effort, they reflect the complexity of real inspection workflows and are necessary for evaluating inspection-relevant VL capabilities.

Furthermore, benchmark datasets should draw on data from multiple bridges with consecutive inspections, enabling evaluation of component-level documentation, damage re-identification, and damage progression across time. Figure 6 presents a high-level conceptual organization of an inspection dataset fulfilling this requirement.

A further implication concerns quantitative reasoning tasks, particularly the reading and interpretation of measurements in images. Even in general VLM research, this

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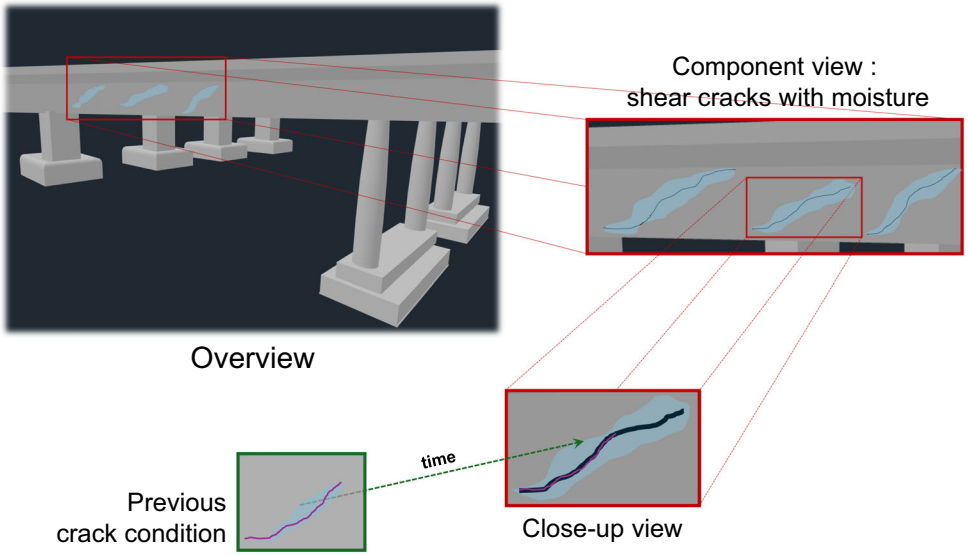


FIGURE 5 Illustration of multi-scale and temporal information required for reliable damage documentation. Example use case: Load-induced shear cracks at a bridge girder. The overview image supports localization of the damage within the overall structure, the component-level view enables interpretation of the cracks as a damage pattern with a common cause rather than isolated defects, and the close-up view provides detailed information on individual cracks. The temporal axis illustrates the same damage in a previous inspection, enabling assessment of crack progression.

```
Bridge
└─ Inspection
   └─ Component
      └─ Damage
         ├─ Damage Type
         ├─ Damage Cause
         ├─ Damage Condition Rating
         ├─ Visual Evidence (Image Set / Video)
         │  └─ VL Annotations
         │     ├─ Captions
         │     ├─ VQA
         │     └─ Visual Grounding
         ├─ Temporal Links
         │  ├─ Previous Observations
         │  └─ Subsequent Observations
         └─ Maintenance Action
```

FIGURE 6 Conceptual organization of inspection dataset for VL benchmarking.

challenge has only recently begun to be addressed.[71,72] Fine-grained visual perception poses a substantial challenge in this context. Regarding bridge inspections, this concerns identifying measurement instruments, correctly localizing measurement marks, and extracting associated values and units. Progress in this area requires a benchmark dataset that explicitly includes inspection images containing measurement tools and reference markers, reflecting realistic documentation practices.

Lastly, inspection-oriented VL benchmarks require consistent evaluation metrics. Commonly used VLM metrics should be critically examined with respect to their suitability for bridge inspection contexts and, where necessary, adapted to inspection-specific language and documentation tasks.

7 | OUTLOOK

This outlook consolidates the technological-conceptual gaps identified in the literature and outlines research directions and objectives for advancing VLM-based bridge inspection beyond the current early exploratory stage highlighted by this review. Addressing these research objectives would help align VLM-based approaches more closely with real-world inspection requirements.

A persistent bottleneck for ML in bridge inspection remains the lack of comprehensive benchmark datasets. In this context, BMS records offer a unique and largely untapped resource: decades of paired visual and textual records, even though they often lack consistent multi-scale image coverage. Future research should therefore focus on curating open-access benchmark datasets derived from BMS records. Stronger collaboration between researchers and road authorities is essential to achieve this goal.

The recording of complex, inspection-critical details requires considerably more attention. While VL foundation models demonstrate strong summarization and interpretation abilities in initial tests, their ability to

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identify subtle attributes, such as distinguishing between wet and dry efflorescence, remains untested. Moreover, recent analyses of German BMS data[62] indicate that missing objects (e.g., missing traffic signs, safety nuts, paving stones) constitute the most frequent damage category. Such cases require reliable anomaly detection,[73] yet current VLM research rarely evaluates models along this dimension. Future studies should therefore investigate whether VLMs can reliably recognize inspection-critical visual details and missing-object anomalies.

Certain damage types require not only textual description but also spatial localization and quantification. Phrase-grounding VLMs with segmentation capabilities have not yet been systematically assessed for bridge inspection, despite the importance of segmentation for documenting damage extent, extracting measurements, and enabling spatio-temporal comparisons across inspections.[74] A closely related, but long-term, research frontier is image-based absolute depth estimation, which, in combination with segmentation, could provide the basis for damage measurement. Recent depth models, such as Depth Anything v2[75] and Marigold,[76] show promising general performance, yet their behavior in the case of concrete damage remains unknown. Extending VLMs to jointly output segmentation masks, textual descriptions, and depth estimates could integrate documentation and measurement tasks. Such integration could reduce manual effort during fieldwork while also supporting more accurate damage quantification, for example by estimating spalling volume beyond simplified width-height-depth approximations. This represents an important research objective for future studies.

Beyond perceptual capabilities, reliability and consistency of VLM outputs remain central challenges. Given the risk of model errors and hallucinations, ontology-supported ML is gaining traction in recent years.[77] Existing damage ontologies for concrete structures[78,79] and recent attempts to link defect ontologies with LLMs[80] provide important sources. However, substantial ontology extension and comprehensive integration of ontologies with VLMs is required. Embedding such models within digital-twin environments, where structural data, visual inspection data, historic BMS records, sensor data, and (N)DE results are integrated, could improve reliability, particularly for damage cause assessment. This is especially relevant for damage patterns where even experts face substantial uncertainty, such as diagonal cracking in bridges. Therefore, the generation and extension of inspection-relevant ontologies, together with the integration of additional data modalities, represent important long-term research directions.

Future VLM research should explicitly define the technical and operational inspection settings for which VLMs are intended, since factors such as image scale can strongly influence model performance. UAV-based VLM applications are promising in this regard. However, as principal inspections are expected to continue relying on human expertise, understanding and optimizing the interaction between inspectors, UAV systems, and multimodal AI models remains a critical long-term research challenge.

UAV-based video recordings further open new opportunities for advanced applications: VLMs for video captioning could support autonomous navigation, component-wise damage summarization, damage-pattern-based cause inference, and near-real-time report generation. Their practical viability, however, requires thorough validation in real inspection scenarios and is therefore another important research objective to pursue.

As VLMs assume a greater role in inspection workflows, human-AI collaboration (e.g., reference 81) is emerging as a key research area. The influence of human factors[82] at different levels, from individual inspectors to organizational and environmental contexts, introduces new dimensions for analysis in this emerging sociotechnical setting. In particular, over-reliance on ML predictions poses safety risks. Research on explainability and safety mechanisms in ML software, and decision-making psychology within bridge inspection practice is therefore essential to ensure responsible deployment.

Finally, future VLM research in bridge inspection should avoid repeating patterns observed in earlier crack-detection literature, namely narrowly defined tasks, validation on restricted case studies, and assembling ML-models through trial and error with limited conceptual grounding. Without a shift toward system-level thinking, VLM-based studies risk remaining isolated studies with limited practical value. Some studies suggest that their systems could support less-experienced inspectors. Such claims remain speculative in the absence of field studies identifying which aspects of inspection practice present the greatest challenges.

Sustainable progress requires a clear definition of the automation concept, including camera setup, explicit consideration of regulatory and operational requirements, and use of historic BMS data. Only through such system-level integration can VLM research evolve into robust and deployable solutions for bridge inspection.

8 | CONCLUSION

This review has examined the emerging field of VLM applications for bridge inspection and identified the central opportunities and limitations shaping its development.

Although VL foundation models have advanced rapidly since 2021, their application in bridge inspection

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remains at an early stage. Despite methodological diversity, existing studies focus largely on image captioning and simple damage-component associations, with a noticeable reversion to simple damage classification. Consequently, the zero-shot and interpretative capabilities of VLMs remain largely underutilized in current bridge inspection research.

Across the reviewed studies, a consistent pattern emerges: existing VLM-based approaches are primarily driven by data availability and narrowly defined experimental tasks, rather than by the information requirements and operational realities of bridge inspection practice. As a result, critical aspects such as complex real-world damage scenarios, inspection-dependent data capture settings, and the integration of visual-textual reasoning into inspection and digital-twin workflows remain insufficiently embedded or entirely unaddressed. The central limitation of current VLM-based bridge inspection research therefore lies not in model capability per se, but in a fundamental misalignment between real inspection processes and the way inspection tasks are formulated and datasets are designed. This misalignment is manifested most clearly in the absence of comprehensive, realistic, inspection-oriented benchmarks and corresponding problem definitions. Until these foundations are established, VLM-based systems are likely to remain operationally unviable.

To address this gap and provide a basis for systematic progress, this review contributes the concept of an inspection-oriented VL benchmark framework. Importantly, the prerequisites for realizing such benchmarks partially exist: road authorities and BMSs maintain decades of paired image-text inspection records that, if appropriately curated, preprocessed, and structured, can form the basis of realistic and practice-aligned benchmark datasets.

For the future, advancing VLM-based bridge inspection requires a shift away from isolated case studies toward a system-level perspective that explicitly aligns model development with inspection workflows, digital twins, and human-AI collaboration. Progress will depend on coordinated efforts to establish open, large-scale benchmarks and richer representations of damage and context. If these challenges are addressed, VLMs have the potential to support a transition toward more reliable and efficient digital inspection processes, while preserving the inspector's judgment that remains central to infrastructure assessment.

AUTHOR CONTRIBUTIONS

Rona Firdes Çelik: Conceptualization (lead); investigation (lead); visualization (lead); writing (lead). **Ulas Bagci**: Conceptualization (lead); supervision (lead); writing—review and editing (supporting). **Sylvia Kessler**: Funding acquisition (lead); writing—review and editing (supporting).

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CONFLICT OF INTEREST STATEMENT

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

No data were used for this study.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

APPENDIX A: A OVERVIEW ON DATASETS USED IN VLM STUDIES

TABLE A1 Overview of datasets used in the reviewed studies. The table may show inconsistency in wording and information level; we retain the authors' original terminology and report all information as provided.

Paper	Training data
Chun et al.[41]	**Dataset name**: MLIT dataset **Source**: Japanese bridge inspection reports from 3118 bridges in the Kanto region (2004–2018) **Size**: 422,290 image-text pairs **Annotation**: Each image is linked to one or more textual descriptions specifying the damage type and the affected bridge member; images may contain multiple damages **Labels**: 27 damage types and 38 bridge member classes following the Japanese Bridge Periodic Inspection Guidelines (2019) **Additional information**: Includes steel, reinforced concrete, prestressed concrete, and composite bridges **Data availability**: Available for purchase from the Japanese Ministry of Land, Infrastructure, Transport and Tourism (MLIT) via the Nationwide Road Facility Inspection Database
Li et al.[43]	**Dataset name**: Bridge Damage Comprehensive Description (BDCD) **Source**: 447 real bridge inspection reports **Size**: 2876 image-text pairs **Annotation**: Descriptions include component type, damage category, relative spatial position within the bridge, and camera shooting angle **Labels**: 15 damage types and 13 bridge component classes **Data availability**: Available upon request
Chai et al.[44]	**Dataset 1—Bridge Image Captioning (BiCap)** **Source**: Inspection reports from an unspecified infrastructure company **Size**: 391 image-text pairs **Annotation**: Texts were preprocessed to follow a triplet structure (damage type—preposition—component type); numerical information (e.g., "2 meters") was removed; multiple damages or multiple affected components may occur in one image **Labels**: 6 damage types and 25 component types **Dataset 2—Bridge Image Damage Detection (BiDam)** **Source**: Same as BiCap **Size**: Same as BiCap **Annotation**: Bounding-box annotations stored in XML format **Labels**: 6 damage types (severe corrosion, section loss, pitting, missing bolt, hole, corroded drainage spigot) **Data availability**: Available upon request
Wang and El-Gohary[45]	**Source**: Notes recorded by bridge inspectors during inspections, provided by the Washington State Department of Transportation (WSDOT) **Size**: 3000 image-text pairs **Annotation**: Texts were processed to remove (i) location and orientation information, (ii) numerical damage quantities, and (iii) information not visible in the images **Labels**: Not defined; free-form descriptions **Data availability**: Not specified

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TABLE A1 (Continued)

Paper	Training data
Chun et al.[42]	**Dataset**: MLIT dataset[41] **Data availability**: A subset is publicly available at https://drive.google.com/file/d/1zSjJ_DEd2LP5GvTjjAI3ZqVS4tr-aJU2
Ma et al.[49]	**Source**: Bridge inspection reports **Size**: 12,473 image-text pairs **Annotation**: Bounding boxes for object detection with corresponding damage class labels **Labels**: Spalling, crack, efflorescence, scaling, and no_defect **Data availability**: Not specified
Liao and Nakano[50]	**Dataset 1—CODEBRIM** **Source**: Public benchmark dataset[83]; https://github.com/MrtnMndt/meta-learning-CODEBRIM **Size**: 1590 bridge damage images **Annotation**: Bounding boxes for object detection **Labels**: Five damage types (crack, spallation, exposed reinforcement bar, efflorescence, corrosion) **Dataset 2—Dacl10k** **Source**: Public benchmark dataset[84]; https://github.com/phiyodr/dacl10k-toolkit **Size**: 9920 images of bridge components and damage **Annotation**: Semantic segmentation masks **Labels**: 13 damage types (crack, alligator crack, wetspot, efflorescence, rust, rockpocket, hollow areas, cavity, spalling, graffiti, weathering, residual formwork, exposed rebars) and 6 bridge components **Additional information**: Although detailed annotations are provided by the datasets, the reviewed study used only damage-type labels and their textual descriptions as model inputs
Chen et al.[51]	**Public Dataset 1—COCO-Bridge** **Source**: UAV-based dataset[85] **Size**: 774 images of structural bridge details **Annotation**: Bounding boxes for object detection **Labels**: Bridge details including bearing, cover plate termination, out-of-plane stiffener, and gusset plate connection **Public Dataset 2—Labeled Cracks in the Wild (LCW)** **Source**: Public UAV-based dataset[86] **Size**: 3817 crack images **Annotation**: Semantic segmentation of cracks **Labels**: Crack **Public Dataset 3—CCSSSD** **Source**: Bridge inspection reports[86] **Size**: 440 corrosion images **Annotation**: Semantic segmentation of corrosion **Labels**: Four corrosion condition states (good, fair, poor, severe) **New Dataset—Bridge Member Dataset (BMD)** **Source**: Not specified **Size**: 150 images **Annotation**: Semantic segmentation masks **Labels**: Four bridge components (pier, girder, railing, pavement) **Data availability**: Not specified
Yamane et al.[55]	**Source**: MLIT dataset—augmented with 20,079 UAV images **Size**: 442,035 image-text pairs **Annotation**: Text-based annotations in the form of QA pairs; 3,875,010 QA pairs in total (8.8 questions per image on average) covering bridge member type, damage type, and yes/no questions **Labels**: 38 bridge member types and 22 damage types **Data availability**: Not specified
Kunlamai et al.[57]	**Source**: MLIT dataset **Datasets**: Japanese MLIT bridge inspection dataset and its UAV-augmented extension **Scale, annotation, and labels**: Same as Yamane et al.[55]
Zhang and Liu[59]	**Dataset 1—Road Damage Dataset (RDD2022)** **Source**: Public multinational benchmark dataset[87]

(Continues)

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TABLE A1 (Continued)

Paper	Training data
	Size: 45,788 images captured by high-resolution vehicle-mounted cameras **Annotation**: Bounding boxes for object detection **Labels**: Pavement distress types including transverse cracks, longitudinal cracks, alligator cracks, and potholes **Data availability**: Publicly available **Dataset 2** **Source**: Not specified **Size**: 4 image-text pairs **Annotation**: Textual descriptions of pavement distress **Labels**: Detailed crack information including location, size in pixels, type, and safety risk assessment **Data availability**: Available upon request **Dataset 3** **Source**: Not specified **Size**: 285 image-text pairs **Annotation**: Textual descriptions of pavement distress **Labels**: Descriptions specify location (in pixels), size of distress, necessity of repair, and absence of other damage types **Data availability**: Available upon request
Wang et al.[53]	**Dataset 1** **Source**: Public dataset[88] **Size**: 1290 crack images and 508 non-crack images **Annotation**: Semantic segmentation masks **Labels**: Steel cracks **Data availability**: Publicly available **Dataset 2** **Source**: Public dataset[89]; https://github.com/hzlbbfrog/Civil-dataset **Size**: 65 crack images and 50 background images **Annotation**: Semantic segmentation masks **Labels**: Steel cracks **Data availability**: Publicly available
Darji et al.[60]	**Source**: US Federal Highway Administration (FHWA) bridge database **Size**: Five non-destructive evaluation contour maps generated using ground-penetrating radar (GPR), electrical resistivity (ER), impact echo (IE), and ultrasonic surface waves (USW) **Annotation**: None

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